

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

CS/IT 19-202-0303/19-204-0303 DISCRETE COMPUTATIONAL STRUCTURES
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Use logical notation to define and reason mathematically about the fundamental data types and structures used in computer algorithms.
- CO2: Summarise mathematical notations and concepts in discrete mathematics that is essential for computing.
- CO3: Construct proofs using direct proof, proof by contraposition, proof by contradiction and proof by resolution and by mathematical induction.
- CO4: Familiarise mathematical reasoning and proof strategies.
- CO5: Identify and apply the counting principle.
- CO6: Apply graph theory to solve real world problems.
- CO7: Interpret the conceptual background needed to be able to identify structures of algebraic nature, and discover, prove and use properties about them.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

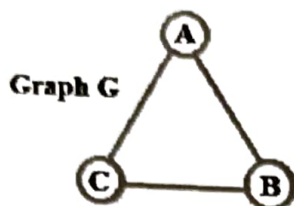
PART A(Answer **ALL** questions)

- | | (8 × 3 = 24) | Marks | BL | CO | PO |
|--|--------------|-------|----|----|-----|
| I. (a) Show that $(p \rightarrow q)$ is logically equivalent to $(\neg p \vee q)$ using truth table. | 3 | 3 | L1 | 2 | 1,2 |
| (b) Translate the following statements in proposition logic: | 3 | 3 | L2 | 3 | 1,2 |
| (i) If you study, you will get good marks. If you do not study, you will enjoy. Therefore, either you will get good marks or you will enjoy. | | | | | |
| (ii) If the catalogue is correct, then if the seeds are planted on April, flowers will bloom in July. | | | | | |
| (iii) If John is elected class representative, then either Mary is elected treasurer or Alice is elected vice-treasurer. | | | | | |
| (c) State Pigeon-hole Principle. How many people must you have to guarantee that at least 9 of them will have birthdays in the same day of the week? | 3 | 3 | L2 | 5 | 1,2 |
| (d) Select a theta notation for each expression and code segment. | 3 | 3 | L3 | 2 | 1 |
| (i) $(4n+1)^2$ | | | | | |
| (ii) $(5n+3)(1+\log n)$ | | | | | |
| (iii) for $i=1$ to $3n$
for $j=1$ to n
$y=y+1$ | | | | | |
| (e) Draw a 3 regular graph with 4 vertices. | 3 | 3 | L2 | 6 | 1 |

(P.T.O.)

- (f) Draw the spanning tree of the given graph.

Marks	BL	CO	PO
3	L2	6	1,2



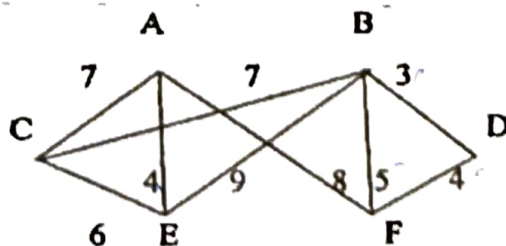
- (g) Define semigroup and monoids with examples.
 (h) Explain the properties of a lattice.

3	L1	7	1
3	L1	7	1

PART B

(4 × 12 = 48)

- II. (a) Identify the domain and range of the relation.
 $R = \{(1,3), (2,2), (3,1), (4,0)\}$
- (b) If $A = \{x, y, z\}$, $B = \{1, 2, 3\}$, $C = \{x, y\}$, $D = \{2, 3\}$ and the relations R and S are defined as $R: A \rightarrow B = \{(x,1), (x,2), (y,3)\}$, $S: C \rightarrow D = \{(x,2), (y,3)\}$, Find $R \cap S$, $R \cup S$, $R \circ S$, R^{-1} .
- (c) Determine whether the relation $R = \{(a, b) \in R : a + b \text{ is even}\}$ on the set of positive integers, is an equivalence relation.
- OR**
- III. (a) Determine the validity of the following argument without using truth tables. "Either I will pass the examination, or, I will not graduate. If I do not graduate, I will go to Australia. I failed: Thus I will go to Australia."
- (b) If $f, g, h : R \rightarrow R$ are defined by $f(x) = x^3 - 4x$, $g(x) = 1/(x^2 + 1)$ and $h(x) = x^4$, find $\{(f \circ g) \circ h\}(x)$ and $\{f \circ (g \circ h)\}(x)$ and check if they are equal.
- IV. (a) Evaluate Big-oh notation for the factorial function and the logarithm of the factorial function, where the factorial function $f(n) = n!$ is defined by $n! = 1 * 2 * 3 * \dots * n$ where n is a positive integer.
- (b) Formulate recurrence relation which can define the number of subsets of an n element set.
- OR**
- V. (a) Write the algorithm to perform bubble sort. Calculate its time complexity in terms of best case and worst case scenarios.
- (b) Solve recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 0$ with initial condition $a_0 = 0$ and $a_1 = 6$.
- VI. (a) Define a bipartite and complete bipartite graph with example.
- (b) Find the minimum spanning tree using Kruskal's algorithm for the given graph

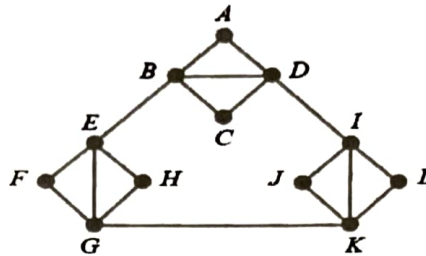


OR

(Continued)

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- | | Marks | BL | CO | PO |
|--|-------|----|----|-----|
| VII. (a) Give an example of a graph that has an Euler circuit which is also Hamiltonian. | 4 | L4 | 6 | 1,2 |
| (b) Write the Fleury's algorithm and apply the same for the following graph. | 8 | L3 | 6 | 1,2 |



- | | | | | |
|---|---|----|---|-----|
| VIII. (a) Explain sublattice, isomorphic lattice and distributive lattice using suitable figures. | 6 | L2 | 7 | 1 |
| (b) Draw the Hasse diagram representing the partial ordering $\{(a, b) a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. | 6 | L2 | 7 | 1,2 |
| OR | | | | |
| IX. (a) Consider an algebraic system $(G, *)$ where G is the set of all non-zero real numbers and $*$ is a binary operation defined by $a*b = ab/2$. Show that G is abelian. | 5 | L3 | 7 | 1,2 |
| (b) Let D be the set of divisors of 60 and divisibility $/$ be the partial ordering. Draw the Hasse diagram. Find the following | 7 | L4 | 7 | 1,2 |
| (i) Upperbound, lowerbound, LUB and GLB of $S = \{12, 30\}$ | | | | |
| (ii) Upperbound, lowerbound, LUB and GLB of $S = \{10, 15\}$ | | | | |

Bloom's Taxonomy Levels

L1 = 12.5%, L2 = 28.3%, L3 = 32.5%, L4 = 26.7%.
